

Chapter 1

Theoretical Background

1.1 Optical Dipole Trap

Optical traps are experimental tools that allow the confinement of atom or ions within a localized region of space, under the condition that these particles have a kinetic energy lower than the trap potential depth (which is usually measured in Kelvin). For the work presented in this thesis, among all possible trapping techniques, the focus will be strictly placed on Optical Dipole Traps (ODTs) for neutral atoms

Optical dipole traps rely on the electric dipole interaction of the atoms with far-detuned light. Following the work of Grimm et al. [1], this section reviews the theory behind the working mechanism of this technique, particularly in the case of red-detuned ODTs.

1.1.1 Oscillator model

The optical dipole force arises from the interaction of the induced atomic momentum with the intensity gradient of the light field. It is a conservative force, thus in the following its potential energy will be derived modelling the atom as an oscillator subjected to a classical oscillating electric field.

1.1.1.1 Interaction between induced dipole moment and driving field

The laser field is modeled as the oscillating electric field:

$$\mathbf{E}(\mathbf{r}, t) = \hat{\mathbf{e}}\mathcal{E}(r)e^{-i\omega t} + c.c. , \quad (1.1)$$

where $\hat{\mathbf{e}}$ is the unitary polarization vector and $\mathcal{E}(r)$ is the complex amplitude of the field. Denoting the atomic polarizability as α , the induced dipole momentum of the atom is given by:

$$\mathbf{p}(\mathbf{r}, t) = \alpha\mathbf{E}(\mathbf{r}, t) = \hat{\mathbf{e}}\alpha\mathcal{E}e^{-i\omega t} + c.c. \quad (1.2)$$

In the semiclassical framework, the interaction of a neutral atom with a laser light field can be described using the dipole approximation. The interaction Hamiltonian takes the form $H_{\text{dip}} = -\mathbf{p} \cdot \mathbf{E}$. For an induced dipole, where the dipole moment depends linearly on the external field, the potential energy is obtained by integrating the interaction over the charging of the dipole:

$$U_{\text{dip}}(t) = \int_0^{\mathbf{E}} -\alpha \mathbf{E}' \cdot d\mathbf{E}' = -\frac{1}{2}\alpha E^2 = -\frac{1}{2}\mathbf{p} \cdot \mathbf{E}. \quad (1.3)$$

Averaging over a field period [1]:

$$U_{\text{dip}} = -2\langle \mathbf{p} \cdot \mathbf{E} \rangle = -\frac{1}{2\epsilon_0 c} \text{Re}[\alpha] I, \quad (1.4)$$

where $I = 1/2\epsilon_0 c |\mathcal{E}|^2$ is the time-averaged field intensity.

This result shows that an atom in an oscillating laser field experiences a force proportional to the field intensity I , and to the real part of the polarizability which represent the in-phase component of the dipole oscillation, responsible for the dispersive properties of the interaction.

It follows that the dipole force is a conservative force proportional to the electric field intensity gradient:

$$\mathbf{F}_{\text{dip}}(\mathbf{r}) = -\nabla U_{\text{dip}} = \frac{1}{2\epsilon_0 c} \text{Re}[\alpha] \nabla I(\mathbf{r}). \quad (1.5)$$

The dipole oscillation absorbs a power

$$P_{\text{abs}} = \langle \dot{\mathbf{p}} \mathbf{E} \rangle = 2\omega \text{Im}[\alpha \mathcal{E}^2] = \frac{\omega}{2\epsilon_0 c} \text{Im}[\alpha] I, \quad (1.6)$$

which is proportional to the out-of-phase component of the oscillating dipole. The absorbed power can be seen as series of absorption and spontaneous emission of photons with energy $\hbar\omega$. Therefore, the total scattering rate is given by

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{P_{\text{abs}}}{\hbar\omega} = \frac{1}{\hbar\epsilon_0 c} \text{Im}[\alpha] I(\mathbf{r}). \quad (1.7)$$

1.1.1.2 Atomic polarizability

The polarizability α can be calculated by considering the classical Lorentz oscillator model for the atom, according to which the polarizability can be calculated by solving the differential equation

$$\ddot{x} + \Gamma(\omega)\dot{x} + \omega_0^2 x = -\frac{eE(t)}{m_e}, \quad (1.8)$$

Using $E(t) = \mathcal{E}_0 \exp(-i\omega t)$, thus looking for a solution in the form $x(t) = x_0 \exp(-i\omega t)$, equation (1.8) gives

$$x_0 = -\frac{e\mathcal{E}_0}{m_e} \frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}. \quad (1.9)$$

Using $p = -ex(t) = -ex_0 \exp(-i\omega t)$ and (1.9), one can easily obtain the polarizability

$$\alpha(\omega) = \frac{e^2}{m_e} \frac{1}{\omega_0^2 - \omega^2 - i\omega\Gamma(\omega)}, \quad (1.10)$$

where $\Gamma(\omega)$ is the classical damping rate due to the radiative energy loss.

$$\Gamma^{\text{class.}}(\omega) = \frac{e^2\omega^2}{6\pi\epsilon_0 m_e c^3}. \quad (1.11)$$

Defining the on-resonance spontaneous emission rate $\Gamma \equiv \Gamma^{\text{class.}}(\omega_0) = (\omega_0/\omega)^2 \Gamma^{\text{class.}}(\omega)$, the polarizability becomes

$$\alpha(\omega) = 6\pi\epsilon_0 c^3 \frac{\Gamma/\omega_0^2}{\omega_0^2 - \omega^2 - i(\omega^3/\omega_0^2)\Gamma}. \quad (1.12)$$

In a quantum two level system subjected to classical radiation, Γ should be substituted with the spontaneous emission rate

$$\Gamma = \frac{\omega_0^3}{3\pi\epsilon_0 \hbar c^3} |\langle e | \hat{d} | g \rangle|^2, \quad (1.13)$$

where $\hat{d}$ is the dipole moment operator.

Optical dipole traps works in the regime of far-detuned low intensity light, thus there is no need to calculate the quantum result, and use the classical result instead [1].

1.1.1.3 Dipole potential and scattering rate

Substituting α form (1.12) in (1.4) and (1.7) one gets the expressions of the dipole potential and the scattering rate of a dipole trap as explicit function of ω , ω_0 and Γ :

$$U_{\text{dip}}(\mathbf{r}) = -\frac{3\pi c^2}{2\omega_0^3} \left(\frac{\Gamma}{\omega_0 - \omega} + \frac{\Gamma}{\omega_0 + \omega} \right) I(\mathbf{r}), \quad (1.14)$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3} \left(\frac{\omega}{\omega_0} \right)^3 \left(\frac{\Gamma}{\omega_0 - \omega} + \frac{\Gamma}{\omega_0 + \omega} \right)^2 I(\mathbf{r}). \quad (1.15)$$

These equations show two resonance frequency for $\omega = \pm\omega_0$. Introducing the detuning $\Delta = \omega - \omega_0$, for the far-red regime it holds that $\Gamma \ll |\Delta|$. For instance, the D_2 line in Rb atoms has a transition frequency of ≈ 3.8 THz thus a trap works generally at few THz of detuning, while $\Gamma \approx 36$ MHz. This allows the introduction of the well-known rotating-

wave approximation, according to which the anti-resonant term can be neglected, further simplifying the equations to

$$U_{\text{dip}}(\mathbf{r}) = \frac{3\pi c^2}{2\omega_0^3} \frac{\Gamma}{\Delta} I(\mathbf{r}) \quad (1.16)$$

$$\Gamma_{\text{sc}}(\mathbf{r}) = \frac{3\pi c^2}{2\hbar\omega_0^3} \left(\frac{\Gamma}{\Delta} \right)^2 I(\mathbf{r}) = \frac{\Gamma}{\hbar\Delta} U_{\text{dip}} \quad (1.17)$$

Equations (1.16) and (1.17) are the fundamental equations governing the behavior of optical dipole traps and yield important physical considerations on the practical use of the optical dipole trap. First the sign of U_{dip} is determined by the sign of the detuning Δ . For red detuning ($\Delta < 0$) the potential is negative, confining atoms at intensity maxima, whereas for blue detuning ($\Delta > 0$), the potential is positive pushing atoms to intensity minima.

1.2 Optical Fibers